

Luca Lazzaroni

Assistant Professor in Electronics

DITEN – UNIGE

BASED IN Genoa, Italy

EMAIL luca.lazzaroni@unige.it

SECTOR RTD-A · IINF-01/A

BIBLIOMETRICS — SCOPUS

68	343	11	21
DOCUMENTS	CITATIONS	H-INDEX	ARTICLES, 5 YRS

SYNCED OCT 19, 2026

01 Profile

Assistant Professor (RTD-A) at the Department of Naval, Electrical, Electronic and Telecommunications Engineering (DITEN), University of Genoa, sector IINF-01/A – Electronics, since March 2025. PhD in Science and Technology for Electronic and Telecommunication Engineering since 2024.

Research on embedded machine learning and edge AI: TinyML toolchains, quantized and binary neural networks on microcontrollers and FPGAs, perception and decision-making for automated driving, deep reinforcement learning, on-device LLMs. The work is also carried out within European and national research projects with industrial partners.

Edge AI · TinyML · Embedded deep learning · Automated driving · Deep reinforcement learning · On-device LLMs

02 Contact & profiles

EMAIL luca.lazzaroni@unige.it	UNIGE rubrica.unige.it	ORCID 0000-0001-8092-5473
SCOPUS 57220892898	GOOGLE SCHOLAR w_6a6sYAAAAJ	GITHUB @lucalazzaroni
LINKEDIN luca-lazzaroni		

03 Appointments & education

POSITIONS

Mar 2025 – present

Assistant Professor (RTD-A)

DITEN, University of Genoa

Scientific sector IINF-01/A — Electronics.

Oct 2023 — Feb 2025	**Research Fellow** DITEN, University of Genoa Grant: “Machine Learning models for the processing of data from autonomous vehicles”.
Nov 2020 — Oct 2023	**PhD Candidate** DITEN, University of Genoa — XXXVI cycle Scholarship: “Deep learning on the edge”.

EDUCATION

2024	**PhD in Science and Technology for Electronic and Telecommunication Engineering** University of Genoa — awarded March 14, 2024 Thesis: “The Facets of Edge AI in Automotive: Exploring Embedded Frameworks, Voice Assistants, and Deep Reinforcement Learning”
2020	**MSc in Electronic Engineering (LM-29)** University of Genoa — July 24, 2020 Thesis: “Design and implementation of a multiplatform execution engine for IoT applications” · Grade: 110/110
2017	**BSc in Electronic Engineering and Information Technologies (L-8)** University of Genoa — October 27, 2017 Thesis: “Model-based design of Lego robot control in Simulink environment”
2014	**Scientific High School Diploma** Liceo Scientifico Statale “Arturo Issel”, Finale Ligure (SV) Grade: 100/100

Licensed Information Engineer — State Examination passed in the second session of 2020.

04 Teaching & academic appointments

COURSES

A.Y. 2024/25 – 2026/27	**Distributed Electronic Systems and Technologies for Environmental Monitoring** Course lecturer · cod. 94631
A.Y. 2024/25 – 2026/27	**Tiny Machine Learning** Co-lecturer · cod. 114752
A.Y. 2026/27	**Electronic Devices and Circuits** Course lecturer · cod. 65997
since A.Y. 2022/23	**Machine Learning for Automated Driving** Subject expert · cod. 108716
A.Y. 2023/24	**Computer Graphics Arch. — Games and Simulation** Adjunct professor · cod. 172317

A.Y. 2022/23	**Autonomous Agents in Games** Teaching support · cod. 198216
A.Y. 2020/21	**Fundamentals of Programming of Electronic Systems** Teaching support · cod. 166061
A.Y. 2020/21	**Makers Approach to Electronic Design** Teaching support · cod. 186985

All courses at the University of Genoa.

SUPERVISION

1 PhD candidate — Ali Zare, Science and Technology for Electronic and Telecommunication Engineering (XLI cycle)

12 BSc theses — Electronic Engineering and Information Technologies (L-8)

10 MSc theses — Electronic Engineering (LM-29)

7 MSc theses — Technological Engineering for Strategy and Security (LM/DS)

2 MSc theses — Computer Science (LM-18)

ACADEMIC APPOINTMENTS

Member of the Teaching and Admissions Committee, MSc in Engineering for Natural Risk Management.

Communications officer for the MSc in Electronic Engineering.

05 Editorial & service

EDITORIAL ROLES

2026 — **Volume Editor**
APPLEPIES 2026 — Springer LNEE
Proceedings in preparation.

2025 — **Editorial Board Member (Early Career)**
Sensors — MDPI
Since December 18, 2025.

2025 — **Volume Editor**
APPLEPIES 2025 — Springer LNEE
Proceedings of the International Conference on Applications in Electronics Pervading Industry, Environment and Society.

2024 — **Guest Editor**
Energies — MDPI
Special Issue “Tiny Machine Learning for Energy Applications”, April 2024 — August 2025.

CONFERENCE ORGANIZATION

Program Chair
APPLEPIES 2026
Bologna, Italy, September 10–11, 2026.

Program Chair
APPLEPIES 2025
Turin, Italy, September 11–12, 2025.

Program Committee
APPLEPIES
Since 2023.

Program Committee
GALA — Games and Learning Alliance
Since 2023.

Session Chair
APPLEPIES 2023
1st Short Oral session — Genoa, Italy, September 28–29, 2023.

PEER REVIEW

JOURNALS

IEEE Access	since 2023
Journal of Systems Architecture	since 2023
Engineering Applications of Artificial Intelligence	since 2024
Sensors	since 2024
IET Intelligent Transport Systems	since 2025
IEEE Transactions on Intelligent Vehicles	since 2026
IEEE Transactions on Circuits and Systems II: Express Briefs	since 2026
Robotics and Autonomous Systems	since 2026
Scientific Reports	since 2026
Journal of Big Data	since 2026
Discover Internet of Things	since 2026

CONFERENCES

IEEE ICRA — International Conference on Robotics and Automation	since 2026
IEEE/RSJ IROS — Intelligent Robots and Systems	since 2025
GALA — Games and Learning Alliance	since 2023
APPLEPIES	since 2021

06 Invited & conference talks

13 TALKS

Sep 11–12, 2025	**APPLEPIES 2025** Turin, Italy
Jan 11–14, 2025	**43rd IEEE ICCE — International Conference on Consumer Electronics** Las Vegas, NV, USA
Sep 19–20, 2024	**APPLEPIES 2024** Turin, Italy
Jun 24–26, 2024	**55th Annual Meeting of the Italian Society of Electronics (SIE)** Genoa, Italy
Sep 28–29, 2023	**APPLEPIES 2023** Genoa, Italy
Sep 7–9, 2023	**54th Annual Meeting of the Italian Society of Electronics (SIE)** Noto (RG), Italy
Sep 26–27, 2022	**APPLEPIES 2022** Genoa, Italy
Sep 7–9, 2022	**53rd Annual Meeting of the Italian Society of Electronics (SIE)** Pizzo (VV), Italy
Sep 21–22, 2021	**APPLEPIES 2021** Pisa, Italy
Dec 1–2, 2021	**GALA 2021 — Games and Learning Alliance** La Spezia, Italy (online)
Jul 7–9, 2021	**52nd Annual Meeting of the Italian Society of Electronics (SIE)** Trieste, Italy (online)
Dec 9–10, 2020	**GALA 2020 — Games and Learning Alliance** Laval, France (online)
Nov 19–20, 2020	**APPLEPIES 2020** Online

07 Awards & memberships

2025

Best Presentation of the Session — “Edge Computing”

43rd IEEE International Conference on Consumer Electronics (ICCE), Las Vegas, NV, USA, January 11–14, 2025

A. Dabbous, L. Lazzaroni, F. Sakr, R. Berta, M. Fresta, F. Bellotti — “Binary Neural Networks for Real-Time Structural Health Monitoring on Edge Devices”

PROFESSIONAL MEMBERSHIPS

IEEE

IEEE Member and IEEE Young Professional.

IEEE Consumer Technology Society

Member of the “Machine Learning, Deep Learning and AI in CE (MDA)” Technical Committee.

SIE — Italian Society of Electronics

Member within the University of Genoa Research Unit since 2021, research area “Electronic systems and applications”.

08 Research projects

Synergies

Generation of scenarios for development, training, virtual testing and validation of CCAM systems

European — HORIZON-CL5-2023-D6-01-02 · Task Leader, WP8 — Evaluation Framework (T8.1)

Generates test scenarios for training, testing and validating CCAM systems, focusing on urban and rural traffic — where relevant-scenario knowledge is far thinner than for highway driving. Runs with most major European carmakers and universities active in the field.

Hi-Drive

Addressing challenges toward the deployment of higher automation

European — H2020, GA 101006664 · Member of the University of Genoa research unit

Develops and tests advanced automated driving at scale, moving from conditional automation (SAE L3) to high automation across different scenarios and road contexts.

L3Pilot

Piloting Automated Driving on European Roads

European — H2020 · Member of the University of Genoa research unit

Large-scale piloting of SAE L3 automated driving functions on public roads across Europe, with the major European carmakers and suppliers.

GEMINI

Green Machine Learning for the IoT

National — PRIN 2022 · Member of the University of Genoa research unit (DII–UNITN / DITEN–UNIGE)

A framework for fully sustainable TinyML applications deployed along an edge-to-cloud continuum, targeting battery-free embedded devices — including those with no network connectivity at all.

ADINA

Integrated Digital Assistant Automotive

Regional — POR FESR Piedmont 2014-2020 · Member of the University of Genoa research unit, with Lifetouch

An in-vehicle virtual assistant on dedicated hardware, connected to the car's systems: AI algorithms interpret the driver's requests and act on vehicle and infotainment functions.

09 Publications

74 ITEMS · 68 IN SCOPUS

2026

Cross-platform deployment and characterization of an on-device generative conversational agent
P. Firpo, R. Berta, F. Bellotti, **L. Lazzaroni**, M. Fresta, A. Pighetti, A. Abedinia, N. Yoonesi, D.P. Pau
Journal of Systems Architecture vol. 181, art. 103981 Journal

A Deployment-Oriented Simulation Framework for Deep Learning-Based Lane Change Prediction
L. Forneris, R. Berta, M. Fresta, **L. Lazzaroni**, H. Rojhan, C. Oh, A. Pighetti, H. Ballout, F. Tango, F. Bellotti
IEEE Signal Processing Letters vol. 33, pp. 136–140 Journal

Efficient Certified Robustness Assessment for Transformer Models via Verifier-Aware Perturbation-Position Scheduling
V. Soltanmuradov, R. Berta, **L. Lazzaroni**, D.M. Gómez, A. Pighetti, F. Bellotti
Machine Learning and Knowledge Extraction vol. 8, no. 9, art. 290 Journal

FPGA Deployment of Extremely Quantized Models for Time-to-Lane-Change Prediction
A. Saad, R. Berta, M. Fresta, **L. Lazzaroni**, A. Pighetti, O. Srour, F. Tango, F. Bellotti
IEEE Transactions on Circuits and Systems II: Express Briefs vol. 73, no. 8, pp. 888–892 Journal

Usability, Technicality, Interoperability (UTI), a Comprehensive Assessment Framework for Digital Platforms
H. Rojhan, F. Bellotti, M. Fresta, **L. Lazzaroni**, R. Berta
IEEE Access vol. 14, pp. 44886–44903 Journal

StabilEdge, Formal Verification of Edge AI Applications
L. Lazzaroni, R. Berta, V. Soltanmuradov, D. Martin Gomez, F. Bellotti
IEEE Access vol. 14, pp. 37416–37434 Journal

Resource-Adaptive FPGA-Based Kernels for Accelerating Self-attention in Transformers
A. Saad, F. Bellotti, H. Ballout, M. Fresta, **L. Lazzaroni**, O. Srour, R. Berta
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 321–327 Conference

Preface
M. Ruo Roch, R. Berta, A. Gómez, **L. Lazzaroni**, M. Martina, P.M. Ros, N. Pazos Escudero
Lecture Notes in Electrical Engineering pp. v–vi Editorial

Optimizing Binary AutoEncoders on STM32 Microcontrollers
H. Ballout, R. Berta, **L. Lazzaroni**, F. Sakr, A. Pighetti, M. Fresta, F. Bellotti
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 301–307 Conference

Modeling Power Electronics in Simulink. A Tutorial for MSc Electronic Engineering Students
L. Motta, F. Bellotti, R. Berta, **L. Lazzaroni**, M. Fresta, N. Busi
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 242–248 Conference

Managing Performance Indicator (PI) Forms Through a State-of-the-Art Big Data Platform
H. Rojhan, F. Bellotti, **L. Lazzaroni**, M. Fresta, L. Forneris, A. Pighetti, R. Berta
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 218–223 Conference

Lightweight Deep Learning for SOC Estimation of Various Lithium-Ion Batteries on Xilinx ZCU104 FPGA
D. Faryar, R. Berta, M. Fresta, A. Saad, **L. Lazzaroni**, H. Ballout, O. Srour, F. Bellotti
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 25–31 Conference

Improving Lane Change Performance in CARLA Through Curvature-Adaptive B-Spline Planning
O.L. Audi, F. Bellotti, R. Berta, Z. Liu, **L. Lazzaroni**, A. Pilla, H. Rojhan, L. Forneris
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 212–217 Conference

Implementing YoloV8 on Zynq UltraScale+MPSoC FPGA Boards
H. Ballout, R. Berta, O. Srour, **L. Lazzaroni**, M. Fresta, A. Saad, F. Bellotti
Proceedings of SIE 2025 Lecture Notes in Electrical Engineering pp. 289–294 Conference

Generation of Synthetic Datasets of Highway Driving Scenarios
M. Cossu, R. Berta, **L. Lazzaroni**, L. Forneris, M. Fresta, A. Pighetti, A. Pilla, J.L. Sauvaget, H. Touil, L. Durville, T. Griffon, G.B. Nejma, F. Hadj Selem, F. Bellotti
IEEE Transactions on Intelligent Transportation Systems Journal

FPGA Implementation of a Robust Decision-Making System for Low-Speed Maneuvering in Automated Driving
A. Saad, H. Ballout, F. Bellotti, M. Fresta, **L. Lazzaroni**, O. Srour, R. Berta
Proceedings of SIE 2025 Lecture Notes in Electrical Engineering pp. 295–301 Conference

Exploring Neural Architecture Search (NAS) for Binary Neural Networks (BNNs) in the New WakeVision
A. Pighetti, F. Bellotti, S. Mohammady, **L. Lazzaroni**, H. Ballout, R. Berta
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 85–90 Conference

Embedded Deployment of an LLM-Based Voice Assistant
L. Lazzaroni, F. Bellotti, A. Pighetti, A. Saad, O. Srour, R. Berta
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 273–278 Conference

Comparing Complete and Incomplete Formal Verification in Deep Learning-Based Object Detection
V. Soltanmuradov, F. Bellotti, R. Berta, D.M. Gómez, **L. Lazzaroni**
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 286–291 Conference

A Dual Branch Quantized Neural Network for Road Surface Classification Using Smartphone Sensors
S.G. Fashkhami, R. Berta, A. Pighetti, M. Fresta, **L. Lazzaroni**, H. Ballout, H. Rojhan, F. Bellotti
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 183–190 Conference

A Collaborative GitHub Framework for Project-Driven Learning in Microcontroller-Based Embedded Systems
E. Castagno, F. Bellotti, R. Berta, **L. Lazzaroni**, M. Fresta, S. Incerti
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 249–254 Conference

A CARLA Parking Environment for Training Deep-Reinforcement Learning Agents
A. Zare, F. Bellotti, R. Berta, **L. Lazzaroni**, L. Forneris, A. Pighetti
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 79–84 Conference

2025

An Embedded System Architecture for Robust Deep Learning-Based Traffic Light Detection
L. Lazzaroni, F. Bellotti, A. Pighetti, L. Forneris, M. Fresta, H. Ballout, C. Oh, S. Pantawane, R. Berta
IEEE Sensors Journal vol. 25, no. 24, pp. 44859–44874 Journal

TinyML Acceleration with MAX78000
A. Dabbous, **L. Lazzaroni**, F. Bellotti, S.M. Presentación, A. Pighetti, R. Berta
Proceedings of SIE 2024 Lecture Notes in Electrical Engineering pp. 468–474 Conference

Setting Up a Lightweight Simulation Environment for Automated Driving Dataset Collection
L. Forneris, F. Bellotti, R. Berta, M. Cossu, M. Fresta, R. Rezaieamale, H. Rojhan, V. Soltanmuradov, F. Tango, **L. Lazzaroni**
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 156–163 Conference

Runtime Input Monitoring of a Reinforcement Learning-Based Agent for Automated Car Parking
H. Ballout, R. Berta, I. Bochenko, **L. Lazzaroni**, A. Dhonthi, V. Hashemi, F. Bellotti
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 387–394 Conference

Robustness Verification of a Reinforcement Learning-Based Agent for Automated Car Parking
F. Bellotti, R. Berta, V. Soltanmuradov, D.M. Gómez, A. Dhonthi, V. Hashemi, **L. Lazzaroni**
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 139–147 Conference

RL-Based Generation of a Synthetic Automotive Driving Scenario Dataset
R. Berta, F. Bellotti, M. Cossu, L. Forneris, **L. Lazzaroni**, A. Pighetti
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 395–402 Conference

RAFT: Regularized Adversarial Fine-Tuning to Enhance Deep Reinforcement Learning for Self-Parking
A. Pighetti, F. Bellotti, R. Berta, A. Cavallaro, **L. Lazzaroni**, C. Oh
IEEE Sensors Letters vol. 9, no. 9, art. 6010204 Journal

Quantization of MobileNetV2 for Resource-Constrained Microcontrollers
L. Lazzaroni, F. Bellotti, A. Dabbous, A. Pighetti, R. Berta
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 497–504 Conference

Performance Comparison of YOLOv8 and YOLOv10 for Traffic Light Detection on a Jetson Nano Board
H. Ballout, R. Berta, A. Dabbous, M. Fresta, **L. Lazzaroni**, A. Pighetti, F. Bellotti
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 242–249 Conference

On-Board Deployability of a Deep Learning-Based System for Distraction and Inattention Detection
M. Fresta, F. Bellotti, **L. Lazzaroni**, H. Ballout, A. Pighetti, Ó. Ameneiro-Prieto, M. Porta-Lorenzo, D. Sánchez-Louzán, C. González-Escudeiro, F. Tango, R. Berta
IEEE Open Journal of the Industrial Electronics Society vol. 6, pp. 1351–1369 Journal

Leveraging Neural Architecture Search for Structural Health Monitoring on Resource-Constrained Devices
A. Capello, R. Berta, M. Fresta, **L. Lazzaroni**, F. Bellotti
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 323–330 Conference

Green Machine Learning for the IoT (GEMINI): A Position Paper
L. Lazzaroni, A. Dabbous, M. Nardello, M.A. Lebdeh, R. Berta, K.S. Yildirim, F. Bellotti, D. Brunelli
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 339–346 `Conference`

Exploring Decision Transformer for Highway Automated Driving
L. Forneris, F. Bellotti, R. Berta, **L. Lazzaroni**, C. Oh
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 123–130 `Conference`

Deep Learning-Based Real-Time Driver Cognitive Distraction Detection
M. Fresta, F. Bellotti, I. Bochenko, **L. Lazzaroni**, G. Merlhiot, F. Tango, R. Berta
IEEE Access vol. 13, pp. 26589–26607 `Journal`

Comparing Detection-Based and Motion-Based Preprocessing of Video Frames for Automated Driving Scenario Detection
M. Cossu, R. Berta, M.A. Bagheri Orumi, **L. Lazzaroni**, J.L. Sauvaget, H. Touil, F. Bellotti
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 250–257 `Conference`

Binary Neural Networks for Real-Time Structural Health Monitoring on Edge Devices
A. Dabbous, **L. Lazzaroni**, F. Sakr, R. Berta, M. Fresta, F. Bellotti
2025 IEEE International Conference on Consumer Electronics (ICCE)
Digest of Technical Papers IEEE International Conference on Consumer Electronics `Conference`

Benchmarking Microcontrollers with Ultra-Low Resolution Images Classification
A. Dabbous, R. Berta, **L. Lazzaroni**, A. Pighetti, F. Bellotti
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 234–241 `Conference`

2024

An embedded end-to-end voice assistant
L. Lazzaroni, F. Bellotti, R. Berta
Engineering Applications of Artificial Intelligence vol. 136, art. 108998 `Journal`

Adversarial Policy Generation in Automated Parking
A. Pighetti, F. Bellotti, R. Berta, A. Cavallaro, **L. Lazzaroni**, C. Oh
ECCV 2024 Workshop on Multimodal Perception and Comprehension of Corner Cases in Autonomous Driving `Workshop`

CBin-NN: An Inference Engine for Binarized Neural Networks
F. Sakr, R. Berta, J. Doyle, A. Capello, A. Dabbous, **L. Lazzaroni**, F. Bellotti
Electronics vol. 13, no. 9, art. 1624 `Journal`

End-to-End Dataset Collection System for Sport Activities
M. Fresta, F. Bellotti, A. Capello, A. Dabbous, **L. Lazzaroni**, F. Ansovini, R. Berta
Electronics vol. 13, no. 7, art. 1286 `Journal`

YoloP-Based Pre-processing for Driving Scenario Detection
M. Cossu, R. Berta, L. Forneris, M. Fresta, **L. Lazzaroni**, J.L. Sauvaget, F. Bellotti
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 418–423 `Conference`

Low-Cost, Edge-Cloud, End-to-End System Architecture for Human Activity Data Collection
M. Fresta, A. Dabbous, F. Bellotti, A. Capello, **L. Lazzaroni**, A. Pighetti, R. Berta
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 444–449 Conference

Investigating Adversarial Policy Learning for Robust Agents in Automated Driving Highway Simulations
A. Pighetti, F. Bellotti, C. Oh, **L. Lazzaroni**, L. Forneris, M. Fresta, R. Berta
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 124–129 Conference

Development of Deep-Learning-Based Autonomous Agents for Low-Speed Maneuvering in Unity
R. Berta, **L. Lazzaroni**, A. Capello, M. Cossu, L. Forneris, A. Pighetti, F. Bellotti
Journal of Intelligent and Connected Vehicles vol. 7, no. 3, pp. 229–244 Journal

Developing a TinyML Image Classifier in an Hour
R. Berta, A. Dabbous, **L. Lazzaroni**, D.P. Pau, F. Bellotti
IEEE Open Journal of the Industrial Electronics Society vol. 5, pp. 946–960 Journal

Building a Pipeline for Efficient Production of Synthetic Datasets for Improving RL in Automated Driving
L. Lazzaroni, A. Pighetti, F. Bellotti, R. Berta
Proceedings of SIE 2023 Lecture Notes in Electrical Engineering pp. 351–356 Conference

Bringing Intelligence to the Edge for Structural Health Monitoring: The Case Study of the Z24 Bridge
A. Dabbous, R. Berta, M. Fresta, H. Ballout, **L. Lazzaroni**, F. Bellotti
IEEE Open Journal of the Industrial Electronics Society vol. 5, pp. 781–794 Journal

Automated Parking in CARLA: A Deep Reinforcement Learning-Based Approach
L. Lazzaroni, A. Pighetti, F. Bellotti, A. Capello, M. Cossu, R. Berta
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 352–357 Conference

Application of a Serious Game for Emotion Elicitation Under Socio-Economic and Trust Based Decision-Making Scenarios for Autistic Adolescents
F. Ahmed, R. Berta, F. Bellotti, F. Floris, **L. Lazzaroni**, G. Barresi, J.R. Carrion
Games and Learning Alliance
Lecture Notes in Computer Science Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics
pp. 170–180 Conference

2023

Implementing Deep Reinforcement Learning (DRL)-based Driving Styles for Non-Player Vehicles
L. Forneris, A. Pighetti, **L. Lazzaroni**, F. Bellotti, A. Capello, M. Cossu, R. Berta
International Journal of Serious Games vol. 10, no. 4, pp. 153–170 Journal

Socio-Economic Decision Making and Emotion Elicitation with a Serious Game in the Wild
F. Ahmed, R. Berta, F. Bellotti, **L. Lazzaroni**, F. Floris, G. Barresi, J.R. Carrion
Applied Sciences vol. 13, no. 11, art. 6432 Journal

Supporting a .csv-based Workflow in MongoDB for Data Analysts
M. Fresta, A. Capello, F. Bellotti, **L. Lazzaroni**, M. Cossu, R. Berta
2023 IEEE 32nd International Symposium on Industrial Electronics (ISIE)
IEEE International Symposium on Industrial Electronics Conference

Investigating High-Level Decision Making for Automated Driving
A. Capello, L. Forneris, A. Pighetti, F. Bellotti, **L. Lazzaroni**, M. Cossu, A. De Gloria, R. Berta
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 307–311 `Conference`

Explaining a Deep Reinforcement Learning (DRL)-Based Automated Driving Agent in Highway Simulations
F. Bellotti, **L. Lazzaroni**, A. Capello, M. Cossu, A. De Gloria, R. Berta
IEEE Access vol. 11, pp. 28522–28550 `Journal`

Efficient Uploading of.Csv Datasets into a Non-Relational Database Management System
M. Fresta, F. Bellotti, A. Capello, M. Cossu, **L. Lazzaroni**, A. De Gloria, R. Berta
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 9–15 `Conference`

Developing a Toolchain for Synthetic Driving Scenario Datasets
M. Cossu, R. Berta, A. Capello, A. De Gloria, **L. Lazzaroni**, F. Bellotti
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 222–228 `Conference`

Designing an Interpretability Analysis Framework for Deep Reinforcement Learning (DRL) Agents in Highway Automated Driving Simulation
F. Bellotti, **L. Lazzaroni**, A. Capello, M. Cossu, A. De Gloria, R. Berta
Proceedings of SIE 2022 Lecture Notes in Electrical Engineering pp. 239–244 `Conference`

Deep Reinforcement Learning for Automated Car Parking
L. Lazzaroni, F. Bellotti, A. Capello, M. Cossu, A. De Gloria, R. Berta
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 125–130 `Conference`

2022

Synthetic Driving Scenarios Generator
M. Cossu, R. Berta, A. Capello, A. De Gloria, **L. Lazzaroni**, F. Bellotti
53rd Annual Meeting of the Italian Society of Electronics (SIE), Pizzo (VV) `National`

Investigating High-Level Decision Making for Automated Driving
A. Capello, L. Forneris, A. Pighetti, F. Bellotti, **L. Lazzaroni**, M. Cossu, A. De Gloria, R. Berta
53rd Annual Meeting of the Italian Society of Electronics (SIE), Pizzo (VV) `National`

Explaining a Deep Reinforcement Learning (DRL)-Based Automated Driving Agent in Highway Simulations
F. Bellotti, R. Berta, A. De Gloria, A. Capello, M. Cossu, **L. Lazzaroni**
53rd Annual Meeting of the Italian Society of Electronics (SIE), Pizzo (VV) `National`

An Embedded End-to-End Voice Assistant
L. Lazzaroni, F. Bellotti, A. De Gloria, R. Berta
53rd Annual Meeting of the Italian Society of Electronics (SIE), Pizzo (VV) `National`

Assessing Versatility of a Generic End-to-End Platform for IoT Ecosystem Applications
R. Berta, F. Bellotti, A. De Gloria, **L. Lazzaroni**
Sensors vol. 22, no. 3, art. 713 `Journal`

High-Level Decision-Making Non-player Vehicles
A. Pighetti, L. Forneris, **L. Lazzaroni**, F. Bellotti, A. Capello, M. Cossu, A. De Gloria, R. Berta
Games and Learning Alliance
Lecture Notes in Computer Science Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics
pp. 223–233 `Conference`

Developing a Synthetic Dataset for Driving Scenarios
J. Motta, F. Bellotti, R. Berta, A. Capello, M. Cossu, A. De Gloria, **L. Lazzaroni**, S. Bonora
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 310–316 `Conference`

Classifying Simulated Driving Scenarios from Automated Cars
M. Cossu, J.L.Q. Villon, F. Bellotti, A. Capello, A. De Gloria, **L. Lazzaroni**, R. Berta
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 229–235 `Conference`

A Serious Game for Using Socio-Economic and Trust Based Decision-Making Scenarios for Elicitation of Emotional Responses
F. Ahmed, J.R. Carrion, F. Bellotti, **L. Lazzaroni**, G. Barresi, R. Berta
Games and Learning Alliance
Lecture Notes in Computer Science Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics
pp. 43–52 `Conference`

2021

Project and Development of an Italian Offline Embedded Voice Assistant
L. Lazzaroni, F. Bellotti, A. De Gloria, R. Berta
52nd Annual Meeting of the Italian Society of Electronics (SIE), Trieste `National`

Edgine, A Runtime System for IoT Edge Applications
R. Berta, A. Mazzara, F. Bellotti, A. De Gloria, **L. Lazzaroni**
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 261–266 `Conference`

Adapting Autonomous Agents for Automotive Driving Games
G. Campodonico, F. Bellotti, R. Berta, A. Capello, M. Cossu, A. De Gloria, **L. Lazzaroni**, T. Taccioli, F. Davio
Games and Learning Alliance
Lecture Notes in Computer Science Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics
pp. 101–110 `Conference`

2020

Employing an IoT Framework as a Generic Serious Games Analytics Engine
L. Lazzaroni, A. Mazzara, F. Bellotti, A. De Gloria, R. Berta
Games and Learning Alliance
Lecture Notes in Computer Science Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics
pp. 79–88 `Conference`